Notes

1. INTRODUCTION TO DBMS

Definition

A Database Management System (DBMS) is software used to create, store, manage, and retrieve data in an organized manner. It acts as an interface between users and the database.

Need for DBMS

Before DBMS, data was stored in file systems, which caused:

- Data redundancy (duplicate data)
- Data inconsistency
- Difficult data access
- Lack of security

DBMS solves these issues by:

- organizing data efficiently
- controlling access
- ensuring consistency

Functions of DBMS

- Data storage and organization
- Data retrieval using queries
- Security and authorization
- Backup and recovery
- Transaction management

Advantages of DBMS

- Reduces redundancy
- Improves consistency
- Provides security
- Supports multiple users
- Ensures data integrity

Example

A college database stores student details, courses, and marks in structured tables.

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2. DATABASE CONCEPTS AND TYPES

What is a Database?

A database is a collection of logically related data stored in an organized way.

Basic Concepts

- Data: Raw facts (e.g., numbers, text)
- Information: Processed data
- Table: Data organized in rows and columns

Types of Databases

1. Hierarchical Database

- Data arranged in tree structure
- One parent → many children

2. Network Database

- Multiple relationships between data
- More flexible than hierarchical

3. Relational Database

- Data stored in tables
- Most widely used

Examples: MySQL, PostgreSQL

4. NoSQL Database

- Used for unstructured data
- Flexible schema

Example: MongoDB

Comparison

- Relational → structured
- NoSQL → flexible

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3. DATA MODELS AND ER MODEL

Data Model

A data model defines how data is structured, stored, and accessed.

Types of Data Models

- Hierarchical Model
- Network Model
- Relational Model

ER Model (Entity-Relationship Model)

Used to design database structure visually.

Components

Entity

A real-world object
Example: Student

Attribute

Property of an entity
Example: Name, Age

Relationship

Connection between entities
Example: Student enrolls in Course

Types of Relationships

- One-to-One
- One-to-Many
- Many-to-Many

Example

Student → Enrolls → Course

Importance

- Helps design database clearly
- Reduces errors before implementation

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4. RELATIONAL MODEL AND KEYS

Relational Model

Data is stored in tables (relations).

Each table has:

- Rows (tuples)
- Columns (attributes)

Keys in DBMS

1. Primary Key

- Unique identifier
- Cannot be null

Example: Student_ID

2. Foreign Key

- Links two tables
- Maintains relationship

Example: Course_ID

3. Candidate Key

- Possible primary key

4. Composite Key

- Combination of multiple attributes

Example: (Student_ID, Course_ID)

Importance of Keys

- Avoid duplicate data
- Maintain relationships
- Ensure data accuracy

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5. NORMALIZATION

Definition

Normalization is the process of organizing data to reduce redundancy and improve consistency.

Goals

- Remove duplicate data
- Improve consistency
- Ensure proper dependencies

Normal Forms

First Normal Form (1NF)

- Atomic values
- No repeating groups

Second Normal Form (2NF)

- Must be in 1NF
- No partial dependency

Third Normal Form (3NF)

- Must be in 2NF
- No transitive dependency

Example

Instead of storing everything in one table:

- Create separate tables for students and courses

Advantages

- Reduces redundancy
- Improves performance
- Easier data management

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6. SQL BASICS

Definition

SQL (Structured Query Language) is used to manage and manipulate databases.

Types of SQL Commands

1. DDL (Data Definition Language)

Used to define database structure

- CREATE
- ALTER
- DROP

Example:
CREATE TABLE students (id INT, name VARCHAR(50));

2. DML (Data Manipulation Language)

Used to modify data

- INSERT
- UPDATE
- DELETE

Example:
INSERT INTO students VALUES (1, 'John', 20);

3. DQL (Data Query Language)

Used to retrieve data

- SELECT

Example:
SELECT * FROM students;

4. DCL (Data Control Language)

Used for permissions

- GRANT
- REVOKE

Importance of SQL

- Easy to use
- Powerful data manipulation
- Widely used in industry

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7. SUMMARY

- DBMS is used to manage data efficiently
- Databases store structured information
- ER model helps in design
- Keys maintain relationships
- Normalization reduces redundancy
- SQL is used to interact with databases